THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

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Family Name _____
First Name _____

School of Electrical Engineering & Computer Science

Semester 2 Examinations, 2025

ELEC7309 Power System Planning and Reliability

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•No written or printed material permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

2 x 14-Page Answer Booklets
1 x 1cm x 1cm Graph Paper

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

There are two sections in this exam (PART A and PART B). Answer each section (PART A and PART B) in a separate booklet. This final exam is worth 60 marks, which accounts for 60% out of the full 100% for the course.

For Examiner Use Only

Question	Mark

Total _____

PART A: POWER SYSTEM PLANNING
ANSWER ALL TWO QUESTIONS FROM PART A

Question A1 **(Total 15 marks)**

(a) You work for APD Engineering and have been asked to assess the suitability of investing a new power plant by calculating the Benefit–Cost Ratio (BCR) using discounted cash flow. APD's investment rule: projects with BCR < 1.6 will not be funded.

Using the plant data in Table 1, compute the BCR and state whether the project is feasible (meets the APD threshold) or not feasible. Explain briefly why. **(6 marks)**

Table 1 for Question A1 (a)

Total initial investment ($)	4,000,000
Asset life (Years)	30
Annual operating cost ($)	150,000
Annual gross benefit ($)	700,000
Discount rate (%)	6

(b) Three generation technology options (A, B and C) are under consideration. The capital and operating parameters for each technology are given in Table 2. Assume a discount rate of 7% and that any one of the technology options may be chosen to serve the system.

Table 2 for Question A1 (b)

Technology type	Investment cost ($/kW)	Plant life (Years)	Fuel cost ($/MWh)
A	1,200	30	18
B	500	25	28
C	150	20	45

(1) Derive an expression for the annualised cost (in $/MW) as a function of capacity factor (CF) for each technology using the data given in Table 2.

(4 marks)

(2) Using the screening curve technique, find the optimal generation type and capacity to supply the demand characterised by the Load Duration Curve shown in Figure 1.

(5 marks)

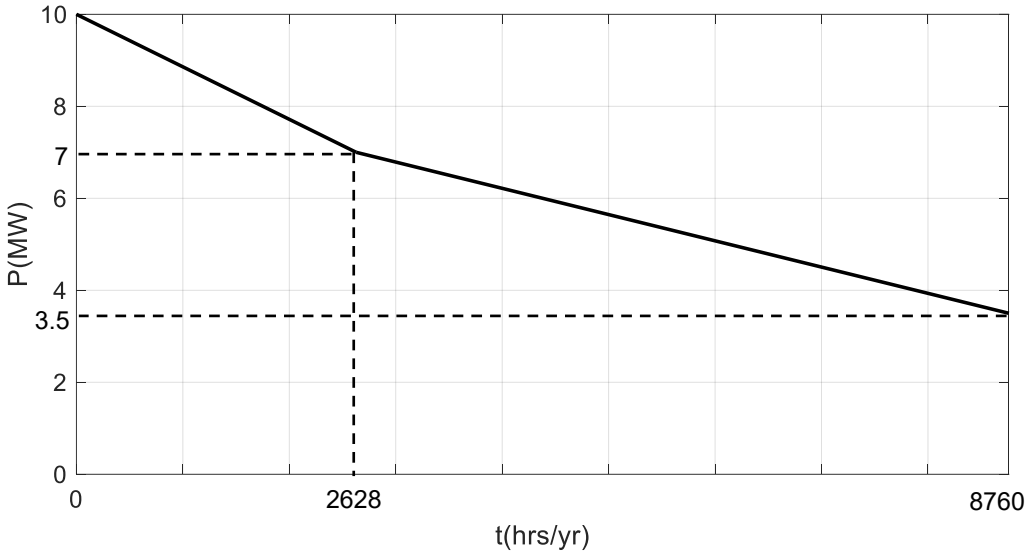


Figure 1 for Question A1 (b): Load duration curve

Question A2 **(Total 15 marks)**

(a) Consider a peer-to-peer (P2P) energy market consisting of six prosumers and six consumers. None of the participants has battery storage. The bidding information for prosumers and consumers is provided in Table 3 and Table 4, respectively. Assume that the price of energy trading in the peer-to-peer market is decided by a double auction technique, and customers are served according to their bidding sequence in the market (first bid, first served). The energy selling price by the grid is 20 cents/kWh, and the feed-in-tariff price is 7 cents/kWh.

Table 3 for Question A2 (a): Bidding of various prosumers

Prosumer ID	**A**	**B**	**C**	**D**	**E**	**F**
Bidding sequence	1	2	3	4	5	6
Price (cents/kWh)	8	10	12	11	13	15
Energy surplus (kWh)	4	5	7	9	9	5

Table 4 for Question A2 (a): Bidding of various consumers

Consumer ID	**P**	**Q**	**R**	**S**	**T**	**U**
Bidding sequence	1	2	3	4	5	6
Price (cents/kWh)	9	10	13	14	8	15
Energy demand (kWh)	3	5	4	7	8	6

(1) Calculate the auction price. **(3 marks)**

(2) Determine the cost to each consumer. **(4 marks)**

(b) Consider the two-bus power system shown in Figure 2. The marginal cost (MC) of production of the generators connected to buses A and B is given by the following expressions.

$$MC_A = 20 + 0.03\,P_A \text{ (\$/MWh)}$$

$$MC_B = 15 + 0.02\,P_B \text{ (\$/MWh)}$$

where P_A and P_B are in MW.

Assume that the demand is constant and insensitive to price, energy is sold at its marginal cost of production and there are no limits on the output of the generators. The transmission line between Buses A and B is in service, with a maximum transfer capacity of 600 MW (i.e., the maximum allowable power flow from Region B to Region A is 600 MW). The output of the generators is unlimited.

(1) Calculate the electricity price in both regions. **(2 marks)**
(2) Calculate the cost of generation production per hour in both regions. **(2 marks)**
(3) Calculate the cost per hour of electricity for customers in both regions. **(2 marks)**
(4) Calculate the generator revenues per hour in both regions. **(2 marks)**



Figure 2 for Question A2 (b)

PART B: POWER SYSTEM RELIABILITY QUESTIONS ARE ON THE NEXT PAGE

PART B: POWER SYSTEM RELIABILITY
ANSWER ALL TWO QUESTIONS FROM PART B – USE A NEW ANSWER BOOK

Question B1 **(Total 15 marks)**

Two interconnected generating systems, System A and System B, are interconnected by one 30 MW tie line.

System A: Contains four 10 MW units (forced outage rate = 2%) and one 20 MW unit (forced outage rate = 2%). The peak load over a 100-day period is 40 MW. Assume a straight-line load curve from 100% to 75% of peak load.

System B: Contains five 20 MW units (forced outage rate = 2%) and one 30 MW unit (forced outage rate = 2%). The peak load over a 100-day period is 90 MW. Assume a straight-line load curve from 100% to 75% of peak load.

System B is assisted by System A under a traditional "assist if possible" interconnection agreement, where System A can supply surplus capacity to System B only when this does not cause a loss of load in System A.

Scenario 1: System B without assistance from System A

(a) Derive the Capacity Outage Probability (COP) table for System B, clearly showing available capacity states, outage capacity states and individual state probabilities. **(3 marks)**

(b) Using the COP table, determine the Loss of Load Expectation (LOLE) in hours/100 days and Loss of Energy Expectation (LOEE) in MWh/100 days of System B for a 100-day period, based on the given load curve. **(4 marks)**

Scenario 2: System B with assistance from System A

(c) Construct the equivalent assisting unit model of System A that can provide support to System B through the 30 MW tie line. **(2 marks)**

(d) Derive the modified COP table of System B with assistance from the equivalent assisting unit of System A and determine the new LOLE (hours/100 days) and LOEE (MWh/100 days) of System B with assistance from the equivalent assisting unit of System A.**(6 marks)**

Question B2 **(Total 15 marks)**

A single-line diagram of a radial three-phase distribution system (Figure 3) includes one upstream circuit breaker at the substation. Assume any fault (single-phase or otherwise) trips all three phases. The failure rate of each line segment is proportional to its length. Protective practice is unchanged: every fault on the main feeder trips the upstream breaker; each lateral has a fuse that clears only the lateral for lateral faults.

Main-feeder sections: failure rate =0.08 f/km/yr, repair time =4 hours.

Lateral distributors: failure rate =0.25 f/km/yr, repair time r=2.5 hours.

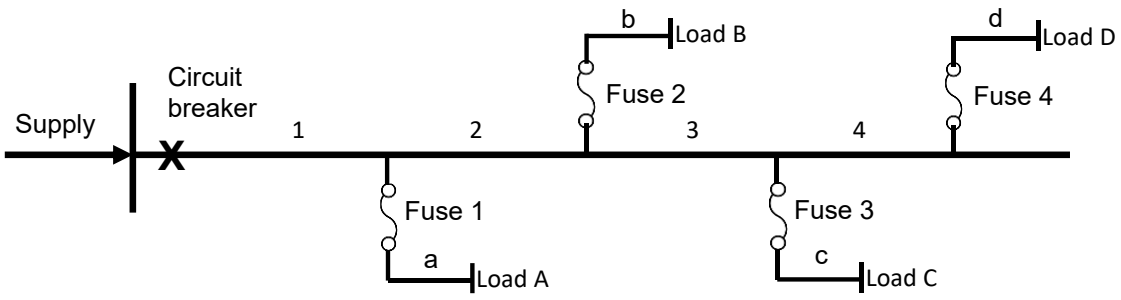


Figure 3 for Question B2

Reliability parameters for the system of Figure 3 are shown in Table 5. Customers and load connected to the system in Figure 3 are shown in Table 6.

Table 5 for Question B2: Reliability Parameters

Component	Length (km)	Component	Length (km)
Section- 1	1.8	Distributor -a	0.8
Section- 2	2.4	Distributor -b	1.6
Section- 3	1.2	Distributor -c	0.9
Section- 4	2	Distributor -d	1.1

Table 6 for Question B2: Customers and Load Details

Load point	Number of customers	Average load connected (kW)
A	380	2,100
B	260	1,450
C	310	1,900
D	210	1,100

Based on these data and the customer/load information, compute the system's reliability indices: SAIFI, SAIDI, CAIDI, AENS and ASAI.

END OF EXAMINATION

FORMULAE NEEDED FOR ANSWERING THE QUESTIONS ARE ON THE FOLLOWING PAGES

FORMULAE – PART A

Variable names and meaning

P_0	Present value
F_n	Value in n years
A	Annuity (constant)
A_n	Annuity in year n
i	Interest rate
B_{jt}	Profit of plan j in year t
C_{jt}	Operation cost of plan j in year t
K_{jt}	Investment of plan j in year t
j	Number of the plan in consideration
n_j	Horizon of study (service/lifetime) of plan j
x_g	Electricity selling price of the grid
x_{FT}	Feed-in-tariff price
x_{MP}	Mid-market price
π	Marginal cost of the generator
P	Power produced by the generator
R	Revenue to the generator
PC	Production cost to the generator
D	Total demand of the customer
J	Cost per hour for customers

Present value

$$P_0 = \frac{F_n}{(1+i)^n}$$

Present Value Factor $(P|F,i,n) = \frac{1}{(1+i)^n}$

Future value

$$F_n = (1+i)^n \,.P_0$$

Future Value Factor $(F|P,i,n) = (1+i)^n$

Constant Annuity

$$F_n = A\frac{(1+i)^n - 1}{i} = A(F|A,i,n)$$

$$A = F_n\frac{i}{(1+i)^n - 1} = F_n(A|F,i,n)$$

$$P_0 = \frac{F_n}{(1+i)^n} = A\frac{(1+i)^n - 1}{i \cdot (1+i)^n} = A\frac{1-(1+i)^{-n}}{i} = A(P|A,i,n)$$

$$A = P_0\frac{i \cdot (1+i)^n}{(1+i)^n - 1} = P_0\frac{i}{1-(1+i)^{-n}} = P_0(A|P,i,n)$$

Capital Recovery Factor

$$CRF_j = \frac{i \cdot (1+i)^{n_j}}{(1+i)^{n_j} - 1} = \frac{i}{1-(1+i)^{-n_j}} = (A|P,i,n_j)$$

Incremental Annuity

$$P_0 = \sum_{k=1}^{n} A_k(1+i)^{-k} = \sum_{k=1}^{n} A_k(P|F,i,k)$$

Net present value

$$NPV_j = \sum_{t=1}^{n_j}\left[B_{jt}(P|F,i,t)\right] - \sum_{t=1}^{n_j}\left[(C_{jt}+K_{jt})(P|F,i,t)\right]$$

Benefit/Cost Ratio

$$BCR_j = \frac{\sum_{t=1}^{n_j}\left[(B_{jt}-C_{jt})(P|F,i,t)\right]}{\sum_{t=1}^{n_j}\left[K_{jt}(P|F,i,t)\right]}$$

Present Value of Costs

$$PVC_j = \sum_{t=1}^{n_j}\left[\left(C_{jt}+K_{jt}\right)(P|F,i,t)\right]$$

Mid-market Price (Price set via mid-market rule)

$$x_{MP} = \frac{x_g + x_{FT}}{2}$$

Capacity Factor (CF)

$$CF = \frac{kWh\ produced\ in\ a\ year}{rated\ power\ (kW) \times 8760h/year}$$

Revenue to generator per hour

$$R(P) = \pi \cdot P$$

Production cost to generator per hour

$$PC(P) = \int_0^{P_B} \pi_B dP$$

Cost per hour for customers

$$J = \pi \cdot D$$

FORMULAE -PART B

Basic Probability Theory

Permutation ${}_nP_r = \frac{n!}{(n-r)!}$

Combinations ${}_nC_r = \frac{{}_nP_r}{r!}$

Mutual inclusive events (independent) P(A∩B)=P(A).P(B)
Mutual inclusive events (dependent) P(A∩B)≠P(A).P(B)
Mutually exclusive events P(A|B)=P(B|A)=0
Complementary events P(A) + P(B) = 1 and P(A|B)=P(B|A)=0
Conditional events P(A|B) = P(A∩B) /P(B), P(B|A) = P(A∩B) /P(A)
Occurrence of at least one of two events (Independent but NOT mutually exclusive events) P(AUB) =P(A)+P(B)-P(A).P(B)
Occurrence of at least one of two events (Independent but mutually exclusive events) P(AUB) =P(A)+P(B)
Occurrence of at least one of two events (Dependent but not mutually exclusive events) P(AUB) =P(A)+P(B)-P(B|A).P(A)=P(A)+P(B)-P(A|B).P(B)

Conditional probability: $P(A) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i)$

Binomial distribution $(p+q)^n = \sum_{r=0}^{n} {}_nC_r p^r q^{n-r} = 1$

Generating Capacity Adequacy

Failure rate/Forced Outage Rate: λ
Repair Rate: μ
Mean Time Before Failure (MTBF) $m = \frac{1}{\lambda}$

Mean Time To Repair (MTTR): $r = \frac{1}{\mu}$

Availability: $A = \frac{\mu}{\mu+\lambda}$

Unavailability: $U = \frac{\lambda}{\mu+\lambda}$

LOLE = t_k x p_k

Here;
t_k = number of time units in the study interval that outage magnitude of Q_k results
p_k = individual probability of capacity outage of Q_k

LOEE= E_k x p_k
E_k = energy curtailed due to capacity outage Q_k

For Composite Power System Reliability Assessment
Recursive algorithm

$$p(X) = p'(X)(1-U) + p'(X-C)U$$

$$\lambda_{-}(X) = \frac{p'(X)(1-U)(\lambda'_{-}(X)+\lambda) + p'(X-C)U(\lambda'_{-}(X-C))}{p(X)}$$

$$\lambda_{+}(X) = \frac{p'(X)(1-U)\lambda'_{+}(X) + p'(X-C)U(\lambda'_{+}(X-C)+\mu)}{p(X)}$$

$$P(X) = P(Y) + p(X)$$

$$F(X) = F(Y) + p(X)\left(\lambda_{+}(X) - \lambda_{-}(X)\right)$$

Here, Y denotes the capacity outage state just larger than X MW
p (X), λ_+ (X) and λ_- (X) are the individual state probability and the upward and downward capacity departure rates, respectively, after the unit is added
For X < C ; p'(X - C) = 0, $\lambda'_+(X-C) = 0$ and $\lambda'_-(X-C) = 0$
For X ≠ 0, C_1, the procedure is initiated with the addition of the first unit (C_1), in this case; $\lambda_+(0) = 0$, $\lambda_-(0) = \lambda_1$, $\lambda_+(C_1) = \mu_1$, $\lambda_-(C_1) = 0$, $\lambda_+(X) = \lambda_-(X) = 0$

$$P_k = \sum_j P(B_j)(P_{gj} + P_{lj} - P_{gj}P_{lj})$$

Here;
B_j= outage state j in the transmission network
P_{gj} = probability of generating capacity outage at state j exceeding the reserve margin
P_{lj} = probability of state j at load bus k exceeding the maximum load that can be supplied at that bus without failure.
P_c = probability of transmission inadequacy

$$F_C = \sum_{j=1}^{2^m} \sum_{i=1;i\neq j}^{2^m} \underbrace{(1-P_C(j))} \cdot \underbrace{P_C(i)} \cdot \lambda_{ji} \qquad P_C = \sum_{j=1}^{2^m} P(j) \cdot P_C(j)$$

$$F = \underbrace{(1-P_G)} \cdot F_C + \underbrace{(1-P_C)} \cdot F_G$$

For Distribution Reliability Assessment
Availability: $A = 1$

Unavailability: $U = \lambda \cdot r = \frac{\lambda}{\mu}$

Common Customer Reliability Indices

System Average Interruption Frequency Index

$$\text{SAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers served}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

System Average Interruption Duration Index

$$\text{SAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customers served}} = \frac{\sum N_i \cdot Q_i}{\sum N_i}$$

Customer Average Interruption Frequency Index

$$\text{CAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers affected}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

Customer Average Interruption Duration Index

$$\text{CAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customer interruptions}} = \frac{\sum N_i \cdot Q_i}{\sum N_i \cdot F_i}$$

Average Service Availability Index

$$\text{ASAI} = \frac{\text{customer hours of available service}}{\text{customer hours demanded}} = \frac{\sum N_i \cdot (1 - Q_i)}{\sum N_i}$$

Average Service Unavailability Index $\text{ASUI} = 1 - \text{ASAI}$

Energy Not Supplied

$$\text{ENS} = \sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}$$

Average Energy Not Supplied

$$\text{AENS} = \frac{\text{total energy not supplied}}{\text{number of customers served}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$

Average Curtailment Index

$$\text{ACCI} = \frac{\text{total energy not supplied}}{\text{number of customers affected}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$